

The Experiment of K-anonymity

- Write the code to implement the privacy-preserving method based on “*K-anonymity*” algorithm
 - You can use Mondrian method, as described in this paper:
[K. LeFevre, D. J. DeWitt, and R. Ramakrishnan, "Mondrian multidimensional k-anonymity," ICDE. Vol. 6. 2006.](#)
- Then, apply “*K-anonymity*” algorithm in your original dataset, to get the result data.
- Then, run some machine learning models (e.g. SVM, Deep Learning) on the above result data & original data, and observe how the privacy protection impacts on the machine learning system.
- Finally, provide the results
 - How do you perturb the data?
 - What is your machine learning model used in this experiment? Besides, provide some details/parameters of this model.
 - Effectiveness measure: Misclassification Rate, Accuracy, Precision, Recall, and AUC
 - Privacy level: the value of K
- Dataset: Adult, or any other dataset
- Programming Language is not limited.

Reference

- Hayden Wimmer, and Loreen Powell. "A comparison of the effects of k-anonymity on machine learning algorithms." Proceedings of the Conference for Information Systems Applied Research ISSN, Vol. 2167. 2014.

Reference

- You can use the popular libraries: “Pytorch”, “Tensorflow”, “Keras”.
- You can also read the below blog posts:
 - <https://machinelearningmastery.com/pytorch-tutorial-develop-deep-learning-models/>
 - <https://medium.com/datadriveninvestor/building-neural-network-using-keras-for-classification-3a3656c726c1>
 - <https://machinelearningmastery.com/tutorial-first-neural-network-python-keras/>

Reference

- The metrics of machine learning
 - [Misclassification Rate](#)
 - [Accuracy, recall, precision, and related metrics](#)
 - [ROC and AUC](#)

Upload the result to Moodle

- Upload your code and report to Moodle.
 - Zipped to a file. The file name should be “Student ID_Program.zip”